

SHAHEER FARDAN

Key AI Engineering Projects

Deep Learning • Generative AI • LLM Systems • Model Optimization • Computer Vision •
Production MLOps

10–11×

LLM throughput

73.7%

model-size reduction

98.97%

clinical sensitivity

92%

ViT classification

Selected flagship work from the full-range AI/ML resume

Portfolio at a Glance

LLM Systems

Inference optimization
Fine-tuning & alignment
RAG

Model Optimization

QAT / PTQ
Pruning / distillation
Graph & hardware optimization

Computer Vision

Classification
Detection / OCR
Segmentation / medical imaging

Generative AI

GAN / DDPM
VAE / WGAN-GP
Synthetic data

Production AI

SageMaker Pipelines
Docker / ECR
Endpoints / APIs / autoscaling

Systems Foundation

Linux / UNIX
HA / clustering / storage
Performance & RCA

Theme: architecture → training → post-training → optimization → serving → production troubleshooting

LLM Inference Optimization — Llama 3.2

10–11×

throughput improvement

92%

lower batch latency

24

controlled experiments

ROUGE-1

quality maintained

Optimization stack

- KV caching for incremental autoregressive decoding
- Structured pruning + quantization
- Speculative decoding implemented from scratch
- DeepSpeed tensor / pipeline parallelism
- Batch inference and latency / throughput benchmarking

Serving pipeline

Prompt

Tokenize

Prefill

Build KV cache

Decode

Incremental
Q/K/V

Optimize

Batch / quant /
speculate

Measure

Latency •
throughput •
quality

Engineering focus: identify the bottleneck, change one variable, benchmark, preserve generation quality.

Mobile AI Model Compression — MobileNetV3

5.96 → 1.57
MB
model size

73.7%
size reduction

6.9×
faster CPU inference

−0.6%
accuracy vs baseline

Techniques evaluated

- Quantization-Aware Training (QAT)
- Graph optimization and operator fusion
- Structured pruning
- Knowledge distillation
- Multi-stage optimization pipelines

Deployment-oriented experiment design

Baseline Measure size • latency • accuracy

Ablations Compare individual techniques

Pipelines Combine complementary optimizations

Final QAT + Graph Optimization

Deploy TorchScript Mobile package

Outcome: explicit mobile constraints met with a measurable accuracy-latency-size trade-off.

Hardware-Aware CNN Optimization — Clinical Deployment

~96%

smaller model

~98%

lower peak memory

~68%

lower latency

98.97%

clinical sensitivity

Architecture redesign

- ResNet-18 redesigned for deployment constraints
- Depthwise-separable convolutions
- Inverted residual blocks
- ONNX graph optimization
- Validated against production SLA requirements

Design principle

Do not optimize the metric in isolation

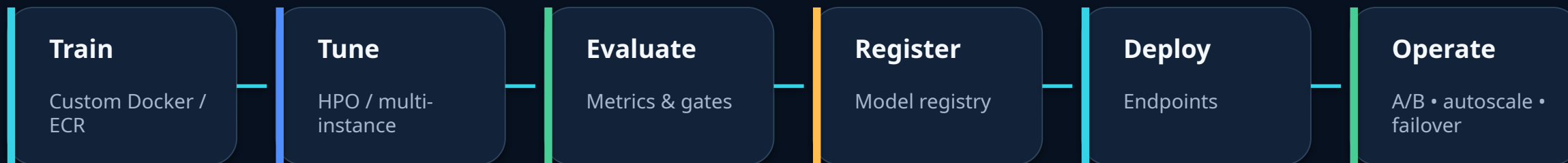
Preserve clinically meaningful sensitivity while reducing model footprint, memory pressure and wall-clock latency.

Hardware-aware architecture

Change the network itself when deployment constraints cannot be solved by post-training optimization alone.

Production MLOps — AWS SageMaker

Repeatable production pattern across ResNet-50 • EfficientNet-B0 • ViT • BERT



Production integration

- Lambda + API Gateway external inference
- Gradio interface for interactive inference
- Multi-model endpoints and deployment patterns

Operational engineering

- Autoscaling and A/B testing
- Failover validation
- Train → evaluate → register → deploy automation

LLM Fine-Tuning, Alignment & RAG

8+

open-source LLMs

**LoRA /
QLoRA**

PEFT

DPO / GRPO

alignment

RAG

grounded generation

Model work

- GPT-2, OPT, BLOOMZ, Phi-2, Qwen, Llama and SmolLM
- Instruction-following and domain-adapted models
- Reward / preference-aligned models
- GSM8K reasoning experiments
- Encoder-decoder translation model from scratch

RAG progression

From fundamentals

Embeddings + cosine-similarity retrieval + grounded context injection

To application

LangChain + Chroma personalized real-estate recommendation assistant

Range signal: post-training quality connected to practical inference and serving constraints.

M.S. Research — Generative AI for Imbalanced Fraud Detection

Synthetic tabular data • class balancing • anomaly detection • hybrid classification

Class balancing + ML

01

SMOTE / ADASYN / VAE / WGAN-GP
Logistic Regression • Random Forest •
XGBoost • ANN

Anomaly detection

02

Autoencoders + Variational Autoencoders
Single and composite anomaly-detection
models

Hybrid CBABCAD

03

Classification + anomaly-detection pipeline
Designed for strong results with model /
balancing-method flexibility

Generative AI emphasis

- VAE and WGAN-GP used to generate synthetic minority-class samples
- Compared statistical fidelity, diversity and downstream ML utility

Research outcome

- Evaluated composite anomaly-detection architectures
- Explored a highly automatable, class-balancing / classifier-agnostic hybrid pipeline

Computer Vision Systems — Detection, OCR & Segmentation

ALPR

Fine-tuned YOLO detector + TrOCR
recognition
Two-stage detect → read pipeline

Traffic-Light Detection

YOLOv8x
52.6% mAP50-75

Retinal Segmentation

Medical image segmentation
77.2 mIoU

Image Classification

ResNet / EfficientNet / ViT
ViT best: 92%

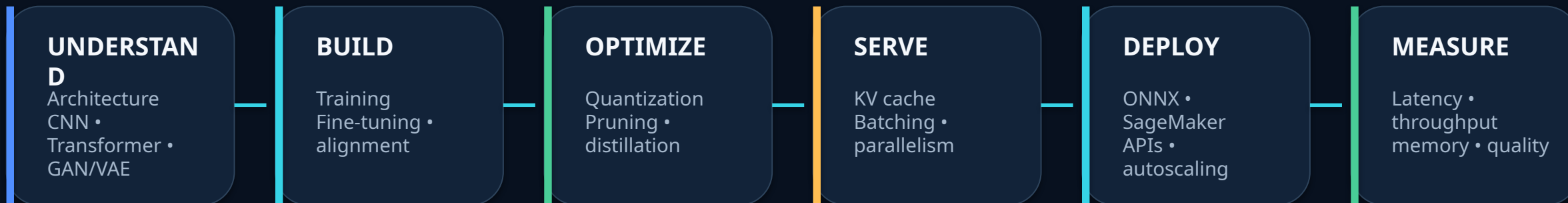
Explainability

Grad-CAM and representation analysis

Generative Vision

DCGAN • Conditional GAN • DDPM • VLM
experiments

How the Projects Connect



Differentiator: model-level understanding + systems-level performance engineering

The portfolio demonstrates an end-to-end progression from architecture and experimentation to optimization, inference, deployment and production troubleshooting.

RESEARCH • OPTIMIZE • DEPLOY • IMPACT

SELECTED AI ENGINEERING PORTFOLIO

Deep Learning / Generative AI

Model Optimization • LLM Systems • Computer Vision • Production MLOps

Architecture → Experimentation → Optimization → Deployment

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